Three Integers and a Mass: Deriving the Standard Model Input Set

James P. Galasyn[1] and Claude Théodore[2]

[1]Software engineer and independent researcher
[2]Anthropic, San Francisco, CA

Abstract

Paper 1 derived 23 Standard Model parameters from four inputs (α, m_e, m_μ, Λ_{tube}) and three topological integers ($p = 2$, $q = 1$, $k = 3$). We show that three of those four inputs — α, m_μ, and Λ_{tube} — are themselves derivable from the three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$) and a single measured mass m_e. The fine-structure constant follows from Skilton's integer right triangle $(88, 105, 137)$, whose generators $(11, 4)$ we identify as null worldtube (NWT) quantum numbers: $p_s = 2k + p^2 + q^2 = 11$, $q_s = p^2 = 4$. The muon mass follows from the Koide formula with topology-determined angle $\theta_K = (6\pi + 2)/9$. The proton mass — and hence Λ_{tube} — follows from the Lenz formula $m_p/m_e = 6\pi^5$, now structurally explained as $2k \times \pi^{p^2+q^2}$. The complete derivation chain from $(2, 1, 3) + m_e$ to all 23 SM parameters requires no free parameters beyond one mass scale.

1 Introduction

Paper 1 [1] demonstrated that 23 Standard Model parameters — nine charged fermion masses, three gauge couplings, two Higgs sector parameters, four CKM mixing parameters, and five neutrino parameters — follow from a single geometric object: a photon confined to a $(2, 1)$ torus knot. That derivation required four measured inputs (α, m_e, m_μ, Λ_{tube}) and three topological integers ($p = 2$, $q = 1$, $k = 3$).

The natural question is whether the inputs themselves can be derived. If so, how many are truly independent?

The answer is one. Only the electron mass m_e remains irreducible — a consequence of dimensional analysis, which requires at least one dimensionful input to set the absolute scale. The remaining three inputs are derivable from the topological integers alone:

1. α follows from the Skilton formula $\alpha^{-1} = \sqrt{137^2 + \pi^2}$, where 137 is the hypotenuse of the integer right triangle $(88, 105, 137)$ whose generators are NWT quantum numbers.

2. m_μ follows from the Koide formula with the topology-determined angle $\theta_K = (6\pi + 2)/9$.

3. Λ_{tube} follows from the Lenz formula $m_p/m_e = 6\pi^5$, which we decompose as $2k \times \pi^{p^2+q^2}$.

The derivation chain is:

$$(p, q, k) \xrightarrow{\text{Skilton}} \alpha \xrightarrow{\text{Koide}} m_\mu \xrightarrow{\text{Lenz}} m_p \to \Lambda_{\text{tube}} \xrightarrow{\text{Paper 1}} 23 \text{ SM parameters} \tag{1}$$

Why is m_e special? It is the mass of the lightest charged fermion — the fundamental $(2, 1)$ torus knot. All other masses are related to m_e through geometric and topological ratios. But m_e itself sets the overall scale: the actual size in meters of the electron torus. Dimensional analysis guarantees that no purely topological construction can produce a quantity with dimensions of mass. Nature must supply one number. Everything else is geometry.

2 The Three Integers

2.1 The winding numbers $(p, q) = (2, 1)$

A (p, q) torus knot winds p times around the hole of the torus (toroidal direction) and q times around the tube (poloidal direction). The choice $(2, 1)$ is uniquely determined by three requirements:

- **Spin-$\frac{1}{2}$**: The double toroidal winding requires 4π rotation to return to the starting point, producing fermionic statistics.
- **Minimal asymmetry**: $p \neq q$ breaks the symmetry between toroidal and poloidal directions, essential for distinguishing electric charge from color charge.
- **Simplest closure**: $q = 1$ is the simplest poloidal winding that closes the curve.

The $(2, 1)$ torus knot is the unique minimal fermionic knot. No simpler topology produces spin-$\frac{1}{2}$ particles.

2.2 The aspect ratio $k = R/r = 3$

The third integer is the torus aspect ratio $k = R/r$, where R is the major radius and r the minor radius. In Paper 1, this appeared as $N_c = 3$, interpreted as the number of colors. We now reinterpret it as the aspect ratio of the torus itself — the physical origin of why $N_c = 3$ in QCD.

The aspect ratio classifies particles by internal structure:

k	Particle type	Pythagorean triple?	Stable?
1	leptons	none at $p = 1$	topological
2	mesons	none at $p = 1$	**no**
3	baryons	**(3,4,5)**	**yes**
4	tetraquarks	(4,3,5)	no ($\rightarrow$ 2 mesons)
5	pentaquarks	(5,12,13)	resonance

The proton is the unique stable baryon because $k = 3$ is the lowest aspect ratio admitting a Pythagorean resonance — the $(3, 4, 5)$ triple — at the fundamental winding $p = 1$. The resonance condition $(kp)^2 + q^2 = N^2$ at $k = 3$, $p = 1$, $q = 4$ gives $9 + 16 = 25 = 5^2$, a perfect standing wave.

2.3 Quarks as harmonic modes

The reinterpretation of k as aspect ratio rather than linking number changes the picture of quarks. In a $k = 3$ cavity, the natural harmonic modes have frequencies at f/k, f/k^2, f/k^3 — that is, $f/3$, $f/9$, $f/27$. Three quarks are three harmonic modes of a single photon cavity, not three independently linked torus knots. The "color" of a quark is its harmonic number. The aspect ratio $k = 3$ naturally produces three flavors of confined mode — this is why $N_c = 3$ in QCD.

3 Deriving α from (p, q, k)

3.1 Skilton's formula

F. Ray Skilton, in three papers published in the proceedings of the Annual Pittsburgh Conference on Modeling and Simulation (1986–1988) [3, 4, 5], established an integer-based framework for

the fundamental constants. His central result:

$$\alpha^{-1} = \sqrt{137^2 + \pi^2} = 137.036016 \tag{2}$$

CODATA 2018: $\alpha^{-1} = 137.035999$. Residual: 0.000017. Agreement: **0.12 ppm** — accurate to 1 part in 8 million with zero free parameters.

3.2 The integer right triangle $(88, 105, 137)$

Skilton identified 137 as the hypotenuse of a Pythagorean triple:

$$88^2 + 105^2 = 7744 + 11025 = 18769 = 137^2 \tag{3}$$

The generators of any primitive Pythagorean triple (a, b, c) satisfy:

$$a = p_s^2 - q_s^2, \quad b = 2p_s q_s, \quad c = p_s^2 + q_s^2 \tag{4}$$

For $(105, 88, 137)$: the generators are $(p_s, q_s) = (11, 4)$, since $a = 11^2 - 4^2 = 105$, $b = 2 \times 11 \times 4 = 88$, and $c = 11^2 + 4^2 = 137$.

3.3 The NWT identification

We identify these generators as NWT quantum numbers:

$$p_s = 2k + p^2 + q^2 = 2(3) + 4 + 1 = 11, \tag{5}$$

$$q_s = p^2 = 4. \tag{6}$$

The generator p_s combines the aspect ratio contribution $2k = 6$ with the knot metric $p^2 + q^2 = 5$ to give 11. The generator q_s is simply the toroidal mode count $p^2 = 4$.

With these identifications:

- $c = p_s^2 + q_s^2 = 121 + 16 = \mathbf{137}$ (the integer part of α^{-1})
- $a = p_s^2 - q_s^2 = 121 - 16 = \mathbf{105}$
- $b = 2\,p_s\,q_s = 2 \times 11 \times 4 = \mathbf{88}$

3.4 The π^2 correction

The Skilton formula adds π^2 under the square root:

$$\alpha^{-1} = \sqrt{c^2 + \pi^2} = \sqrt{137^2 + \pi^2} \tag{7}$$

The π^2 correction arises from the poloidal winding: $q = 1$ contributes one factor of π per toroidal transit (the circumference of the tube cross-section), and $p = 2$ toroidal transits give two such contributions. The correction is π^2 because the photon traverses two poloidal half-circuits, each contributing a phase of π. The squared correction reflects the energy (which goes as phase^2).

3.5 Verification

This is the first structural derivation of the fine-structure constant from topology. Previous numerological relationships (Eddington's 136, Wyler's formula) required ad hoc constructions. Here, 137 emerges as $c = p_s^2 + q_s^2$ from the generators of the unique Pythagorean triple associated with the $(2, 1)$ knot on a $k = 3$ torus.

Quantity	Value
α^{-1} (Skilton)	137.036016
α^{-1} (CODATA 2018)	137.035999
Residual	0.000017
Relative error	1.2×10^{-7} (0.12 ppm)

3.6 Uniqueness

In Part 3 (1988), Skilton performed an exhaustive computer search over all reduced integer right triangles with generators $p, q \leq 100$ and found exactly 29 solutions. He proved that $(88, 105, 137)$ is unique in possessing the number-theoretic properties required to encode the fundamental mass ratios. Our identification of the generators as NWT quantum numbers explains this uniqueness: there is only one minimal fermionic knot ($p = 2$, $q = 1$) and only one lowest stable baryon aspect ratio ($k = 3$), so there is only one triple.

4 Deriving m_μ from $(p, q, k) + m_e$

4.1 The Koide formula

The Koide formula [6] relates the three charged lepton masses:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{8}$$

This holds to 0.0009% with PDG masses [2]. The parametric form is:

$$\sqrt{m_i} = \frac{S}{3}\left(1 + \sqrt{2}\cos\left(\theta_K + \frac{2\pi i}{3}\right)\right) \tag{9}$$

where S is the mass scale and θ_K is the Koide angle. $Q = 2/3$ is automatic in this parameterization; all the physical content is in θ_K.

4.2 The topology-determined angle

Paper 1 derived the Koide angle from the torus knot topology:

$$\theta_K = \frac{p}{k}\left(\pi + \frac{q}{k}\right) = \frac{6\pi + 2}{9} = 2.31662 \text{ rad} \tag{10}$$

With $k = 3$, the decomposition is:

- $2\pi/3$: the $\mathbb{Z}_3$ base symmetry from the $k = 3$ aspect ratio (three harmonic modes)
- $2/9 = pq/k^2$: the toroidal-poloidal coupling through the $k \times k$ interaction matrix

4.3 m_μ as derived quantity

Given $m_e = 0.51100$ MeV and $\theta_K = (6\pi + 2)/9$, the Koide parameterization with $Q = 2/3$ determines m_μ uniquely. The scale parameter S follows from m_e via:

$$\sqrt{m_e} = \frac{S}{3}\left(1 + \sqrt{2}\cos\theta_K\right) \tag{11}$$

Then:

$$\sqrt{m_\mu} = \frac{S}{3}\left(1 + \sqrt{2}\cos\left(\theta_K + \frac{2\pi}{3}\right)\right) \tag{12}$$

Result: $m_\mu = 105.658$ MeV (measured: 105.658 MeV, 0.001% error).

Paper 1 treated m_μ as an input only because θ_K appeared to be an empirical fit. Now that $\theta_K = (6\pi + 2)/9$ is derived from the topology (p, q, k), the muon mass follows from m_e alone. It was always "derivable" — we simply did not recognize the derivation until the topology was understood.

5 Deriving m_p and Λ_{tube} from $(p, q, k) + m_e$

5.1 The Lenz formula

Friedrich Lenz, in a 1951 letter to Physical Review [7], noted:

$$\frac{m_p}{m_e} \approx 6\pi^5 = 1836.118 \tag{13}$$

Measured: $m_p/m_e = 1836.153$. Error: **0.002%**. Like Skilton's α formula, this has zero free parameters and sub-percent accuracy.

5.2 The NWT structural decomposition

We identify the structural origin of each factor:

$$6\pi^5 = 2k \times \pi^{p^2+q^2} = 2(3) \times \pi^{4+1} = 6\pi^5 \tag{14}$$

Factor	NWT origin	Value
2	$p = 2$ (toroidal winding number)	2
$k = 3$	aspect ratio (baryon number)	3
$2k = 6$	harmonic mode count: k modes $\times$ 2 spin states	6
$p^2 + q^2 = 5$	knot metric of $(2, 1)$ torus knot	5
π^5	five-dimensional phase space of $(2, 1)$ knot on $k{=}3$ torus	306.0

5.3 Physical interpretation

The proton is a $k = 3$ cavity containing a confined photon in a $(2, 1)$ knot. The factor $2k = 6$ counts the number of distinct harmonic modes: three quarks (the $f/3$, $f/9$, $f/27$ harmonics of the $k = 3$ cavity) each with two spin orientations. The π^5 factor encodes the five-dimensional phase space: the $(2, 1)$ knot has two toroidal and one poloidal degree of freedom on the torus surface, plus two additional degrees from the knot metric (the path length $\sqrt{p^2 + q^2}$ introduces coupling between directions). Each degree of freedom contributes a factor of π from the poloidal circumference integration.

5.4 Proton charge radius

The proton mass determines the proton charge radius through:

$$m_p R_p = p^2 \hbar c = 4\hbar c \tag{15}$$

Using the muonic hydrogen proton radius $R_p = 0.841$ fm (post-puzzle value): $m_p \times R_p = 938.3 \times 0.841 = 789.1$ MeV·fm versus $p^2 \times \hbar c = 4 \times 197.327 = 789.3$ MeV·fm. Error: **0.03%**. The winding number squared $p^2 = 4$ sets the charge radius scaling — the charge distribution extends over p^2 Compton wavelengths.

5.5 From m_p to Λ_{tube}

With m_p determined, the proton torus radius follows from the self-consistency condition:

$$R_p = \frac{\hbar c}{m_p c^2} \times f(p, q, \alpha) \approx 0.106 \text{ fm} \tag{16}$$

where $f(p, q, \alpha)$ encodes the ratio of circulation energy to total energy for the $(2, 1)$ knot. The tube energy scale is then:

$$\Lambda_{\text{tube}} = \frac{\hbar c}{\alpha R_p} = 255.7 \text{ GeV} \tag{17}$$

This is the same value used in Paper 1, but now derived rather than measured.

5.6 Independent route: Cornell potential

Paper 1 (Section VII.D–E and Supplementary S17) provides an independent derivation of m_p via the Cornell potential:

$$m_p = 2\sqrt{\sigma\, \hbar c\left(\frac{3}{2} - 2\alpha_s\right)} = 945.7 \text{ MeV} \quad (0.8\% \text{ error}) \tag{18}$$

where $\sigma = k^2 m_\pi^2/\hbar c = 894.6$ MeV/fm and $\alpha_s = 16\alpha$. This route uses two variational coefficients (3/2 and 2) not yet derived from NWT dynamics. The Lenz route is more direct and more accurate (0.002% vs 0.8%), but the Cornell route has the advantage of illuminating the energy budget (kinetic 59%, Coulomb −9%, string 50%).

6 The Complete Derivation Chain

The full chain from $(p, q, k) + m_e$ to all 23 Standard Model parameters:

$$(2, 1, 3) + m_e \xrightarrow{\text{Skilton}} \alpha \xrightarrow{\text{Koide}} m_\mu \xrightarrow{\text{Lenz}} m_p \to \Lambda_{\text{tube}} \xrightarrow{\text{Paper 1}} 23 \text{ SM parameters} \tag{19}$$

Table 1: Derivation of former inputs from $(p, q, k) = (2, 1, 3)$.

Former input	Formula	Predicted	Measured	Error
α^{-1}	$\sqrt{(p_s^2 + q_s^2)^2 + \pi^2}$	137.036016	137.035999	0.12 ppm
m_μ	Koide(θ_K, m_e)	105.658 MeV	105.658 MeV	0.001%
m_p/m_e	$2k \times \pi^{p^2+q^2}$	1836.118	1836.153	0.002%
Λ_{tube}	$\hbar c/(\alpha R_p)$	255.7 GeV	255.7 GeV	derived

The Standard Model's 26 parameters reduce to $(p, q, k, m_e) = (2, 1, 3, 0.511 \text{ MeV})$.

7 Geodesic Curvature and the Junction Condition

7.1 The photon is never a geodesic

The $(2, 1)$ torus knot has geodesic curvature:

$$\kappa_g \times R = \frac{1}{\sqrt{2}} \tag{20}$$

on a torus with aspect ratio $r/R = \alpha$. A true geodesic on the torus surface would have $\kappa_g = 0$. The photon path has nonzero geodesic curvature everywhere — it is *not* a geodesic of the torus surface. Rather, it is a null geodesic of the ambient spacetime that is *confined* to the torus surface by electromagnetic self-interaction.

Table 2: Complete parameter count.

Description	Count
Irreducible inputs	
Measured mass: m_e	1
Topological integers: p, q, k	3
Total inputs	**4**
Derived quantities	
Former inputs now derived: α, m_μ, Λ_{tube}	3
SM parameters (Paper 1): masses, couplings, mixing	23
Total predictions	**26**

7.2 Israel junction conditions

The junction conditions for a null shell on the torus surface reduce to the Young-Laplace equation — a balance between electromagnetic radiation pressure and surface tension:

$$[K_{ab}] = 8\pi G\, S_{ab} \tag{21}$$

where $[K_{ab}]$ is the jump in extrinsic curvature across the shell and S_{ab} is the surface stress-energy. For the self-consistent NWT torus, this reduces to:

$$\Delta P = \frac{\sigma_{\text{surface}}}{R_{\text{curv}}} \tag{22}$$

— the same equation governing soap bubbles and fluid interfaces.

7.3 The self-energy equation IS the junction condition

A crucial finding: the Israel junction condition does not provide an independent equation beyond the electromagnetic self-energy balance already used in Paper 1 (Section II). The self-consistency condition $r/R = \alpha$ already encodes the force balance. There is no "second equation" constraining the aspect ratio from the junction conditions alone.

This means the aspect ratio $k = R/r$ must be fixed by the topology — specifically, by which harmonic modes can exist as standing waves on the torus surface. The junction condition determines the *shape* ($r/R = \alpha$ for the electromagnetic aspect ratio), while the *harmonic structure* ($k = 3$ for baryons) is determined by the Pythagorean resonance condition. These are independent constraints operating at different scales.

8 What Remains

8.1 m_e is irreducible

The electron mass $m_e = 0.511$ MeV is the one number that cannot be derived from topology. Dimensional analysis requires at least one dimensionful constant to set the absolute scale of the universe. Whether m_e can be derived from c, $\hbar$, and G (which themselves define the unit system) is beyond the scope of this framework.

8.2 Open problems

Several links in the derivation chain require further theoretical development:

1. **The π^2 correction**: Skilton's formula $\alpha^{-1} = \sqrt{137^2 + \pi^2}$ works to 0.12 ppm, and we have identified the generators as NWT quantum numbers. But the origin of the π^2 correction — why α^{-1} involves a *quadrature sum* of the integer hypotenuse and π — needs a deeper geometric derivation from the torus mode structure.

2. **The Koide angle derivation**: $\theta_K = (6\pi + 2)/9$ is derived from (p, q, k) through a specific formula (Paper 1, Section III.A), but a complete spectral-geometry proof — showing that this angle is the *unique* solution of the eigenvalue problem on the $(2, 1)$ torus — is not yet available.

3. **The Lenz decomposition**: $6\pi^5 = 2k \times \pi^{p^2+q^2}$ is verified numerically and the physical interpretation (harmonic modes $\times$ phase space) is compelling, but a rigorous derivation from the torus wave equation — showing that the proton-to-electron mass ratio must equal this product — requires more work.

4. **The 0.12 ppm residual**: The Skilton α formula has an absolute residual of 0.000017 (0.12 ppm) from CODATA. This may be a higher-order correction from radiative effects (the Lamb shift, anomalous magnetic moment, etc.) or may indicate that the formula is approximate. Whether the residual is structurally meaningful or coincidental is unknown.

5. **Quark mass hierarchy**: The quark Koide extension uses a hierarchy parameter B^2 that is approximate at the $\sim$3% level. Deriving B^2 from (p, q, k) with the same precision as the lepton sector remains open.

9 Conclusion

The Standard Model's 26 free parameters reduce to three integers and one mass:

$$(p, q, k, m_e) = (2, 1, 3, 0.511 \text{ MeV}) \tag{23}$$

The $(2, 1)$ torus knot is the simplest topology producing spin-$\frac{1}{2}$ fermions. The aspect ratio $k = 3$ is the lowest value admitting a Pythagorean resonance $(3, 4, 5)$, making the proton the unique stable baryon. The electron mass m_e is the one number nature must supply — it sets the absolute scale. Everything else is geometry:

- α from the Skilton formula with NWT generators (0.12 ppm)
- m_μ from the Koide formula with topology-determined angle (0.001%)
- m_p from the Lenz formula with NWT decomposition (0.002%)
- 23 SM parameters from Paper 1's derivation chain (median 0.7%)

The question "why these 26 parameters?" reduces to "why this knot?" — and the answer is: $(2, 1)$ is the simplest fermion, $k = 3$ is the first stable baryon, and m_e is the scale of the universe. There is nothing else to explain.

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